Per-Grain Analysis Pipeline: Clustering, Parent-Grain Reconstruction, and Martensite Twin Correspondence

For each region of interest (ROI), corresponding to one reconstructed austenite grain, the following pipeline is applied. ROIs are selected manually, by drawing a polygon around what is judged, from the microstructure map, to correspond to a single pre-deformation austenite grain; this selection is not an objective, automated segmentation and is therefore subject to operator error – an ROI may in reality span more than one original grain, or capture only part of one, and this uncertainty propagates into every subsequent step of the analysis.

1 Misorientation-based sub-grain clustering

Pixels within the ROI are grouped into sub-grains ("clusters") using a misorientation-based region-growing algorithm: neighbouring pixels are merged when their disorientation is below a threshold angle `ang_thr`, subject to a maximum cluster spatial diameter `dmax` and a minimum cluster size `minidxs`. Each resulting cluster's average orientation is subsequently *symmetry-branch consistent* across the whole ROI (`symmetrize_clusters`): since a measured orientation matrix is only defined up to the crystal's point-group symmetry, all clusters' representative orientations are resolved onto a single, mutually consistent symmetric branch before any further analysis.

2 Local misorientation and dislocation-density fields

Pixel-to-pixel boundary misorientations are computed within the ROI. For each pixel, the kernel average misorientation $\langle\theta(x)\rangle$ – the local disorientation averaged over the peripheral pixels at a fixed kernel radius x – is evaluated for a range of kernel radii (1st to nth neighbour shell), following the method of Kamaya [1], as applied to EBSD data by Moussa et al. [2]. Disorientation values above a threshold (`maxmis`) are excluded from the average to prevent grain boundaries from contaminating the local, sub-grain-scale signal. Assuming the disorientation gradient is locally constant, $\langle\theta(x)\rangle$ varies linearly with x; a linear regression over the sampled kernel radii yields both the disorientation gradient $\mathrm{d}\langle\theta\rangle/\mathrm{d}x$ (the slope) and, by extrapolation to $x = 0$, an estimate of the local EBSD measurement noise (the intercept), which would otherwise be conflated with genuine lattice curvature if the GND density were instead estimated directly from $\langle\theta\rangle/x$ at a single kernel radius. The GND density is then obtained from the slope,

$$\rho_{\mathrm{GND}} = \frac{\alpha}{b}\,\frac{\mathrm{d}\langle\theta\rangle}{\mathrm{d}x}, \tag{1}$$

with b the Burgers vector magnitude and α a constant depending on the dislocation character (tilt or twist), evaluated assuming a single slip system, $\langle 100\rangle\{110\}$, in the B2 austenite phase.

3 Homogeneous lattice direction and provisional zone axis

The largest cluster in the ROI (by pixel count) is used as an anchor. For a small set of candidate low-index directions $\{\langle 100\rangle, \langle 110\rangle, \langle 111\rangle\}$, each direction's full crystallographically-equivalent family is tested for how consistently it points along a common sample-frame direction across *every* cluster in the ROI (not only the anchor): for each symmetric variant, the direction is expressed in every cluster's own crystal frame and its angular spread across the pooled cluster set is measured. The variant with the smallest spread is retained as the ROI's provisional common lattice direction ("zone axis"), evidence that the sub-grains share a crystallographic relationship inherited from a common parent.

4 Twin and kink relationship identification

Every pair of clusters in the ROI is tested against a library of tabulated NiTi austenite twinning modes ($\{112\}$–$\{115\}$): for each mode, the full space of symmetry-equivalent branch choices is searched (*T-loop*) together with the mode's tabulated twin elements ($K_1/\eta_1/K_2$), and a pair is accepted as a twin match if the axis and angle of at least one of the K_1-180°, η_1-180°, or K_2-shear conditions falls within a tolerance `tol_deg` of the target. Independently, pairs are also tested for a *kink* relationship: a blind search over low-index candidate axes (up to a maximum Miller index) for any axis about which the pair's misorientation is consistent, with no assumed target angle.

5 Directed parent-orientation reconstruction

The identified twin and kink relationships define a graph over the ROI's clusters. Twin edges are additionally *directed*: for each twin pair, the Schmid factor of the tabulated twin system is evaluated on both sides under the given loading direction `axial_dir`; the side with the higher (more mechanically favourable) Schmid factor is treated as the parent-like side, propagation running only parent→child. Kink edges, for which no independent Schmid factor is defined, remain undirected. A breadth-first traversal from the best-supported source node (a cluster with no incoming directed twin edge, reachable to the largest pixel-weighted subtree) propagates every reachable cluster's orientation back to a common reference frame – using the exact tabulated twin operation for twin edges (denoised relative to the noisy measured relative orientation) and the raw measured relative orientation for kink-only edges – and the reconstructed parent orientation is the pixel-weighted quaternion mean of all these independent votes.

6 Verification against tabulated twin geometry

The reconstructed parent orientation is checked directly against every cluster in the ROI (independently of the graph used to derive it): treating the parent as a hypothetical reference orientation, each real cluster is tested for a twin relationship to it, using the same T-loop/tabulated-mode search as above, with a tolerance set by the recipe's grain-reconstruction threshold. This both confirms the internal consistency of the reconstruction and yields, for every matched cluster, the Schmid factor of its twin system evaluated on the parent side and on the cluster's own side.

7 Matrix identification by orientation merging

Clusters whose true minimum disorientation (searched over the full crystal symmetry group) to the reconstructed parent orientation falls below the same threshold are merged into a single "matrix" cluster, representing the (largely) untwinned parent austenite. The merge additionally records each merged fragment's original disorientation to the reference, for later inspection.

8 Consolidated reconstruction report

For the merged clustering result, every remaining (non-matrix) cluster is re-classified relative to the confirmed matrix orientation: twin relationships (with per-mode breakdown, pixel/cluster counts, and Schmid factors) and, for clusters showing no twin relationship, an additional kink check. This yields a single consolidated report of how the ROI's total pixel population partitions into matrix, twin-related, kink-related, and unidentified fractions.

9 Martensite correspondence and variant selection

If the matrix cluster used for merging exists, it serves as the anchor for a second homogeneous-direction search (restricted to matrix, twin- and kink-matched clusters only), refining the common $\langle 110 \rangle$-type lattice direction shared across the confirmed network; otherwise, the cluster closest in disorientation to the reconstructed parent is used as a fallback anchor. This refined direction is mapped, via the tabulated austenite→martensite direction-correspondence matrices, into martensite Miller indices for each of the twelve correspondence variants; variants whose predicted martensite direction is not parallel to the expected martensite zone axis are discarded. Among the remaining candidate variants, the one giving the maximum (for tensile loading) transformation strain along `axial_dir` is selected as the most probable martensite variant retained prior to reverse transformation. Finally, every confirmed austenite twin in the report is checked for whether its K_1 plane, mapped through the selected variant's correspondence matrix, is consistent with a known, named martensite twinning plane – i.e. whether the austenite twin is itself inherited from a specific martensite twin system.

10 Checkpointing

The clustering result, the merged (matrix-identified) clustering result, the reconstruction report, the martensite-correspondence result, the common lattice direction, the anchor cluster, the loading direction, and the driving recipe are written to a per-grain HDF5 checkpoint, allowing the full downstream analysis and figure generation to be repeated without re-running this pipeline.

References

[1] M. Kamaya, *Assessment of local deformation using EBSD: Quantification of accuracy of measurement and definition of local gradient*, Ultramicroscopy **111** (2011) 1189–1199.

[2] C. Moussa, M. Bernacki, R. Besnard, N. Bozzolo, *Statistical analysis of dislocations and dislocation boundaries from EBSD data*, Ultramicroscopy **179** (2017) 63–72.